

Draft - expert review pending

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Unit 1 assessment

Unit summary

Computer literacy is the full set of skills for using technology effectively: operating devices, handling files, creating documents, finding and judging information, and communicating safely; it is far more than typing. ICT covers computers, phones, networks and the internet together, and runs through education, work and society in Pakistan, where most of the roughly 140 million broadband connections are mobile. Computers evolved from the abacus and mechanical calculators through punched-card machines and the room-sized ENIAC to the personal computer and the smartphone, in five generations defined by vacuum tubes, transistors, integrated circuits, microprocessors and today's artificial-intelligence systems. Computers are classified by functionality (analog, digital, hybrid) and by size and power (supercomputer, mainframe, minicomputer, microcomputer, embedded). Their impact on society is double-edged: instant communication, richer education and new services on one side; the digital divide, misinformation and screen-time concerns on the other. ICT applications run through every major field: education, health, agriculture, banking and government.

Summative assessment

Multiple-choice questions (MCQs)

1. Computer literacy is best defined as: (*Remember*) a) the ability to type quickly b) the knowledge and skill needed to use computers and related technology effectively c) knowing the history of computers d) the ability to repair a computer

2. Saima uses her phone daily for chat and videos but cannot attach a file to an email. This situation shows that: (*Understand*) a) phones are not part of ICT b) computer literacy covers more skills than casual phone use c) email is an outdated tool d) typing speed is the core of computer literacy
3. Which of the following is NOT one of the five DigComp areas of digital competence? (*Remember*) a) Information and data literacy b) Safety c) Communication and collaboration d) Machine repair
4. The technology that defines the first generation of computers is: (*Remember*) a) transistors b) vacuum tubes c) microprocessors d) integrated circuits
5. Jacquard's weaving loom has a place in computing history because it: (*Understand*) a) wove cloth faster than hand looms b) was the first electronic machine c) showed that a machine could follow instructions stored on punched cards d) replaced the abacus in Europe
6. A museum machine from the early 1960s is built from discrete transistors. Its generation is: (*Apply*) a) first b) second c) third d) fourth
7. A bedside device measures a continuous heartbeat signal, converts it into numbers, and raises an alarm when the numbers cross a limit. Its functionality class is: (*Apply*) a) analog b) digital c) hybrid d) it cannot be classified
8. A smartphone is classified as a microcomputer because: (*Understand*) a) it is small enough to fit in a pocket b) its entire processor fits on a single microprocessor chip c) it has no physical keyboard d) it cannot run programs
9. The digital divide is: (*Remember*) a) the speed difference between old and new computers b) the gap between people who have good ICT access and skills and people who do not c) a method of splitting a hard disk d) a type of network cable
10. A grower receives daily market prices by SMS and uses them to decide when to sell. This is an application of ICT in the field of: (*Apply*) a) banking b) government c) agriculture d) entertainment

Restricted-response questions (RRQs)

1. Explain in two sentences why typing speed alone does not make a person computer literate. (*Understand*)
2. Describe any three skills that computer literacy covers, and match each one to a school task that needs it. (*Understand*)
3. A head teacher says: "All my staff are computer literate; every one of them has a smartphone." Analyse what is right and what is missing in this claim. (*Analyze*)
4. Describe what the stored-program idea changed about computers. Two sentences. (*Understand*)
5. Describe the five computer generations, naming the technology that defines each. (*Understand*)
6. A school's oldest computer is an IBM PC from the 1980s; its newest is a tablet running a voice assistant. Compare the two machines by generation, naming the defining technology of each. (*Analyze*)
7. Explain the difference between an analog computer and a digital computer, giving one example of each. (*Understand*)
8. Classify each of the following on the size-and-power axis and justify each answer in one clause: a national weather-forecasting system; the controller inside a washing machine. (*Apply*)
9. Explain the digital divide in two sentences and give one example of it from Pakistan. (*Understand*)
10. A school introduces an online homework portal. Half the pupils submit on time; the rest rarely log in because they share one phone at home. Analyse what this situation shows about ICT and society. (*Analyze*)

Extended-response questions (ERQs)

1. "A teacher in Sindh does not really need computer literacy; the old methods worked fine." Evaluate this claim. Use at least two specific school tasks and one fact about technology access in Pakistan as evidence. (*Evaluate*)
2. Your school wants a "computing history" corner for pupils. Decide which three machines or devices from computing history you would display, and justify each choice by what it teaches about how computers changed. (*Evaluate*)

3. A donation arrives at your school: a desktop computer, a tablet, a biometric attendance machine, an old server rack, and a weather receiver that converts continuous analog signals into digital forecasts. Recommend what the school should deploy and where, and classify each device on both axes as part of your reasoning. (*Analyze*)
4. Analyse the impact of ICT on a school or workplace you know well: describe two things that improved, one challenge that arrived with the technology, and one practical recommendation you would make to the head teacher or manager. (*Analyze*)
5. Drawing on the whole unit (the scope of computer literacy, the kinds of computers a teacher meets, and the impact and applications of ICT in society), evaluate what "computer literacy" should mean for a new teacher in Pakistan today. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. b - computer literacy is the effective use of technology, not one motor skill; option a is the classic misconception this unit corrects.
2. b - casual phone use is a subset of digital skills; attaching a file needs file-handling and communication skills beyond scrolling.
3. d - the five DigComp areas are information and data literacy, communication and collaboration, digital content creation, safety, and problem solving.
4. b - first-generation machines such as ENIAC were built from vacuum tubes.
5. c - the loom's punched cards stored the weaving instructions, the ancestor of stored machine instructions.
6. b - discrete transistors define the second generation (1950s to 1960s).
7. c - it measures an analog signal (continuous heartbeat) and processes it digitally, the definition of a hybrid computer.
8. b - the microcomputer's defining feature is a processor on a single microprocessor chip; size is a consequence, not the definition.
9. b - the divide is about access and skills together, not hardware speed.

10. c - SMS market prices to growers is the agriculture application described in Topic 1.4.

RRQ model answers and mark schemes

1. Model: typing is only one input skill; computer literacy also includes file handling, document creation, finding and judging information, communication and safety, so a fast typist who lacks these is not computer literate. **Marks (2):** 1 for typing being only one skill; 1 for naming at least two other components.
2. Model: any three of operating a device, handling files, creating documents, finding and judging information, communicating, staying safe, each matched to a task such as attendance in a spreadsheet, emailing a file, preparing a notice, judging a website, messaging parents, choosing a strong password. **Marks (6):** 1 per skill correctly named, 1 per matching task.
3. Model: right in that smartphone ownership indicates exposure to ICT; missing that computer literacy is a skills claim, not a device-ownership claim, and daily casual use does not show file handling, document creation, information judgement or safety skills. **Marks (3):** 1 for the valid part; 2 for the analysis of what ownership does not prove.
4. Model: before it, machines were rewired to change the task; after it, instructions lived in the same memory as data, so a machine could load a different program and become a different tool, which is how all computers since have worked. **Marks (2):** 1 for instructions in memory; 1 for the change in how tasks are switched.
5. Model: first, vacuum tubes (ENIAC); second, transistors (IBM 1401); third, integrated circuits (IBM System/360); fourth, microprocessors (IBM PC); fifth, artificial intelligence and very large scale integration (modern AI systems). **Marks (5):** 1 per generation with its technology.
6. Model: the IBM PC is fourth-generation, built on a microprocessor; the tablet is also fourth-generation in construction (microprocessor) but runs fifth-generation-style artificial-intelligence software, so the comparison shows generation is defined by core technology, with the fifth being a shift in use as much as in switches. **Marks (3):** 1 for each machine correctly placed; 1 for naming the defining technology of each; 1 for the comparison point.
7. Model: an analog computer works with continuous physical quantities and measures (slide rule); a digital computer works with discrete 0s and 1s and

counts exactly (laptop). **Marks (3):** 1 for continuous vs discrete; 1 per correct example.

8. Model: the weather-forecasting system is a supercomputer, because national-scale simulation needs the fastest class of machine; the washing-machine controller is an embedded computer, because it is a small computer fixed inside another device running one job. **Marks (4):** 2 per device (1 class, 1 justification).
9. Model: the digital divide is the gap between those with good ICT access and skills and those without; in Pakistan it appears, for example, in rural households whose only computer is one shared phone with a weak signal, while city households have broadband at home. **Marks (3):** 1 for the definition; 2 for a concrete Pakistan example.
10. Model: the portal assumes access (a device, connectivity, skill) that not every pupil has, so the same technology widens existing differences instead of closing them: the divide is not only about owning a device but about how many people must share it and how much time it leaves them; the school needs a fallback (paper option, lab time) so the tool does not create two classes of pupil. **Marks (3):** 1 for identifying the divide at work; 1 for the shared-device analysis; 1 for a realistic implication.

ERQ rubrics

1. **Evaluates the claim (Evaluate).** Criteria: *Position taken and defended (0-2); Uses at least two specific school tasks as evidence (0-3); Uses one accurate fact about technology access in Pakistan (0-2, Analyse: the fact must actually bear on the argument, not decorate it); Recognises what the old methods did well before judging (0-2); Clarity and organisation (0-1).* Total 10.
2. **Chooses three displays and justifies (Evaluate).** Criteria: *Three items chosen from computing history (0-3); Each justification names what the item teaches about change (0-4, Analyse: the justification must connect the item to a change, e.g. punched cards to stored instructions, not just "it is old"); Items span different eras or ideas (0-2); Clarity (0-1).* Total 10.
3. **Deploys the donation (Analyze).** Criteria: *Each device classified on both axes (0-5, 1 per device); Deployment recommendation follows from the classification and a school need (0-3, Analyse: the reasoning must cite the*

device's class or capability); *Practical constraints acknowledged (power, space, connectivity, skills)* (0-1); *Clarity* (0-1). Total 10.

4. **Analyses ICT impact on a real setting (Analyze).** Criteria: *Two improvements described concretely* (0-3); *One challenge identified accurately* (0-2); *Recommendation follows from the evidence presented* (0-3, Analyze: the recommendation must cite the observed evidence, not a generic wish); *Clarity and one-page limit* (0-2). Total 10.
5. **Integrative evaluation (Evaluate).** Criteria: *Draws on at least three parts of the unit (skills scope, generations or kinds, impact or applications)* (0-3); *Each part used accurately* (0-3); *Synthesises into a coherent definition for a new teacher in Pakistan, not a list* (0-3, Analyze: the parts must combine into a judgement); *Clarity* (0-1). Total 10.

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